The University of the West Indies

COMP6940 - Big Data and Data Visualization

Assignment 1 Report

Part 1 - NYC Yellow Taxi (January 2023)

Demand, Pricing, and Temporal Patterns

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Repository	https://github.com/zuhrahmoh/COMP6940__A1
Data Sources	NYC TLC January 2023 Yellow Taxi parquet and taxi zone lookup CSV
Curated Output	`data/curated/part1_taxi_curated.parquet`

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1 Dataset Description

This project analyzes the official NYC Yellow Taxi trip records for **January 2023** together with the TLC taxi zone lookup table. The raw trip file contains **3,066,766** rows, while the lookup table contains **265** rows. Each raw trip record represents one taxi trip and includes pickup and dropoff timestamps, pickup and dropoff zone IDs, passenger count, trip distance, fare amount, tip amount, total amount, and payment information.

The curated dataset remains at the **trip level**: one row corresponds to one cleaned taxi trip. The raw trip data were enriched by joining the taxi zone lookup twice, once for pickup geography and once for dropoff geography, so the curated output includes borough and zone names in addition to the original zone IDs.

The curated dataset includes four groups of variables required by the assignment:

- **Geography**: `PULocationID`, `DOLocationID`, `PU_borough`, `PU_zone`, `DO_borough`, `DO_zone`
- **Time**: `tpep_pickup_datetime`, `tpep_dropoff_datetime`, `pickup_date`, `pickup_hour`, `weekday`, `week_of_year`
- **Core numeric fields**: `passenger_count`, `trip_distance`, `fare_amount`, `tip_amount`, `total_amount`, `payment_type`
- **Engineered fields**: `trip_duration_min`, `speed_mph`, `fare_per_mile`, `tip_rate`, `distance_bucket`

After cleaning, the saved curated dataset contains **2,884,396 rows** and **23 columns**. In Notebook 3, an additional date-scope check was applied for analysis, yielding an analysis dataset of **2,884,354 rows**, with pickup datetimes ranging from **2023-01-01 00:00:05** to **2023-01-31 23:59:59**. This confirms that the final analytical scope matches the required January 2023 time window.

2 Cleaning Rules and Feature Engineering

The cleaned dataset was produced in `02_clean_features.ipynb`. Required engineered fields were created for pickup date, pickup hour, weekday, ISO week number, trip duration, average speed, fare per mile, tip rate, and distance bucket. Safe division logic was used for speed, fare per mile, and tip rate to avoid invalid divisions by zero.

The assignment required six explicit cleaning rules. Table 1 summarizes the before-and-after counts reported by the notebook.

Table 1: Cleaning summary table from Notebook 2

Rule	**Rows before**	**Rows after**	**Rows removed**
Missing pickup/dropoff timestamp	3,066,766	3,066,766	0
Trip duration ≤ 0	3,066,766	3,065,645	1,121
Trip distance ≤ 0	3,065,645	3,020,831	44,814
Passenger count missing or ≤ 0	3,020,831	2,906,607	114,224
Total amount ≤ 0	2,906,607	2,884,397	22,210
Duplicate rows by required key	2,884,397	2,884,396	1

The largest reduction came from invalid or missing passenger counts, followed by non-positive trip distances. This is consistent with the goal of removing trips that would distort downstream metrics such as distance, speed, and fare-based ratios. The cleaned dataset still contains some nulls in geographic lookup fields such as `PU_zone` and `DO_zone`; these arise from unmatched or unknown lookup entries and were retained because the assignment did not require dropping such rows.

3 Required Metrics and Interpretation

3.1 Groupby Metrics

The first set of required metrics used groupby summaries to describe demand, pricing, and tipping patterns.

Daily trip counts by borough and date. Daily demand was calculated for each pickup borough and each pickup date. These daily counts form the base for both descriptive reporting and the later window metrics.

Mean and median fare amount and trip distance by borough and weekday. The weekday summary shows that borough-level travel patterns differ substantially. For example, the Bronx had a Monday mean fare amount of approximately **31.42 USD** and a Monday mean trip distance of approximately **7.38 miles**, reflecting relatively longer trips on average than many Manhattan taxi rides.

Payment type proportions by pickup hour. The notebook computed proportions within each hour rather than globally. For midnight pickups, payment type 1 accounted for approximately **83.0%** of trips, while payment type 2 accounted for approximately **15.7%**, indicating that one payment method is consistently dominant throughout the day.

Mean tip rate by distance bucket. Table 2 summarizes the mean tip rate values reported by the notebook.

Table 2: Mean tip rate by distance bucket

Distance bucket	Mean tip rate
[0,1)	0.119133
[1,3)	0.123695
[3,5)	0.120473
[5,10)	0.112819
[10,+)	0.111042

Short to medium-distance trips, especially in the `[1,3)` bucket, had the highest mean tip rates in the analysis output. Tip rates then decline for longer trips, suggesting that tipping is not strictly proportional to distance.

3.2 Window-Style Metrics

Window-style calculations were then applied to the daily borough counts. The notebook computed:

- a **7-day rolling mean** of daily trip counts for each pickup borough,
- the **day-over-day difference** in daily trips, and
- the **day-over-day percent change** in daily trips.

These metrics make short-term fluctuations easier to interpret by separating raw daily movement from the smoother weekly trend. For example, the first few Bronx entries show a movement from 91 trips on January 1 to 61 on January 2, a day-over-day difference of **-30** and a corresponding negative percent change, before demand rises again later in the week.

3.3 EDA Question 1: Strongest Weekday Effect

The assignment defined weekday effect as:

$$S_{borough} = \frac{\max(\mu_{weekday,borough}) - \min(\mu_{weekday,borough})}{\mu_{borough}}.$$

The notebook found the following values for the analyzed pickup geographies: EWR = 0.8526, Bronx = 0.6363, Brooklyn = 0.4968, Manhattan = 0.3660, Queens = 0.2892, Staten Island = 0.2774, and Unknown = 0.2116. Although EWR has the largest raw ratio, it is not one of the five NYC boroughs and its very small counts make the ratio unstable. Restricting the interpretation to NYC boroughs, the borough with the strongest weekday effect is the **Bronx**, with $S_{borough} = \mathbf{0.6363}$.

This means the Bronx has the largest relative swing between its busiest and least busy weekdays compared with its own average daily demand. In practical terms, weekday differences in Bronx taxi activity are more pronounced than in the other NYC boroughs represented in the dataset.

3.4 EDA Question 2: Anomalous Daily Trip Volumes

Daily total trips were converted to z-scores to identify unusual dates. The two most anomalous dates by absolute z-score are shown in Table 3.

Table 3: Top anomalous dates by absolute z-score

Pickup date	Daily total trips	z-score
2023-01-02	61,903	-2.591726
2023-01-01	70,030	-1.915345

Both dates show substantially fewer trips than the January daily average. The notebook interpretation links these anomalies to the **New Year holiday period**, when commuting patterns and discretionary travel differ from a regular workweek. January 2 has the strongest negative deviation, while January 1 is the second most unusual day.

3.5 EDA Question 3: Pickup Hour vs. Distance Bucket

To compare how strongly `tip_rate` is associated with pickup hour versus distance bucket, the notebook compared the standard deviation of mean tip rates across the two grouping structures. The reported values were:

- standard deviation across **pickup hour** groups: **0.006757**
- standard deviation across **distance bucket** groups: **0.004764**

Because the variation in grouped mean tip rates is larger across pickup hours, the notebook concludes that **tip rate is more strongly associated with pickup hour than with distance bucket**. This supports the interpretation that tipping behavior may depend more on time-of-day travel context, such as commuting windows, airport-related travel, or late-night rides, than on trip length alone.

4 Required Visualizations

This section includes the four required plots from Notebook 3.

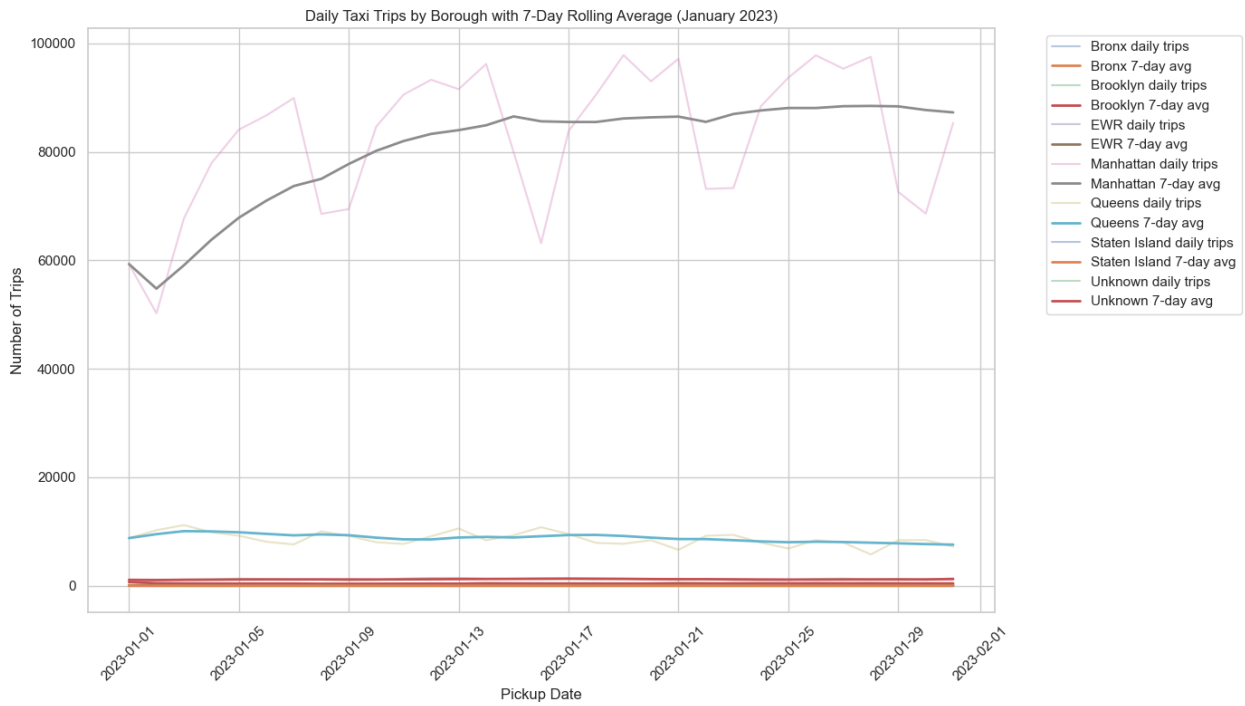

Figure 1: Daily taxi trips by borough with 7-day rolling average (January 2023). Manhattan dominates trip volume throughout the month, while Queens and Brooklyn remain far below Manhattan but above the Bronx and Staten Island. The rolling averages smooth the early-month holiday dip and reveal a more stable demand pattern after the first week.

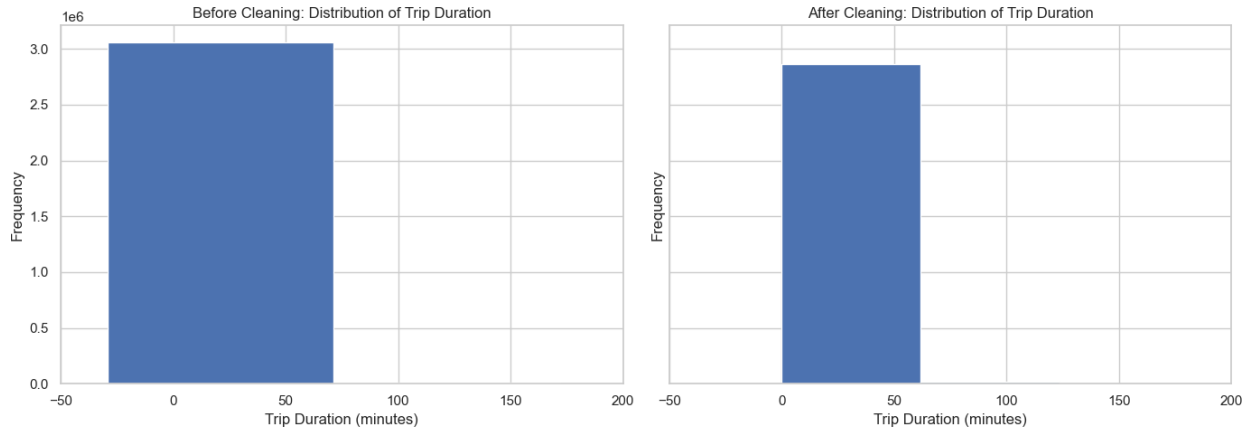

Figure 2: Before-and-after cleaning distribution of trip duration. Before cleaning, the raw data include negative durations and extreme values caused by inconsistent timestamps or erroneous entries. After cleaning, the distribution is restricted to realistic positive trip durations, making downstream summaries more trustworthy.

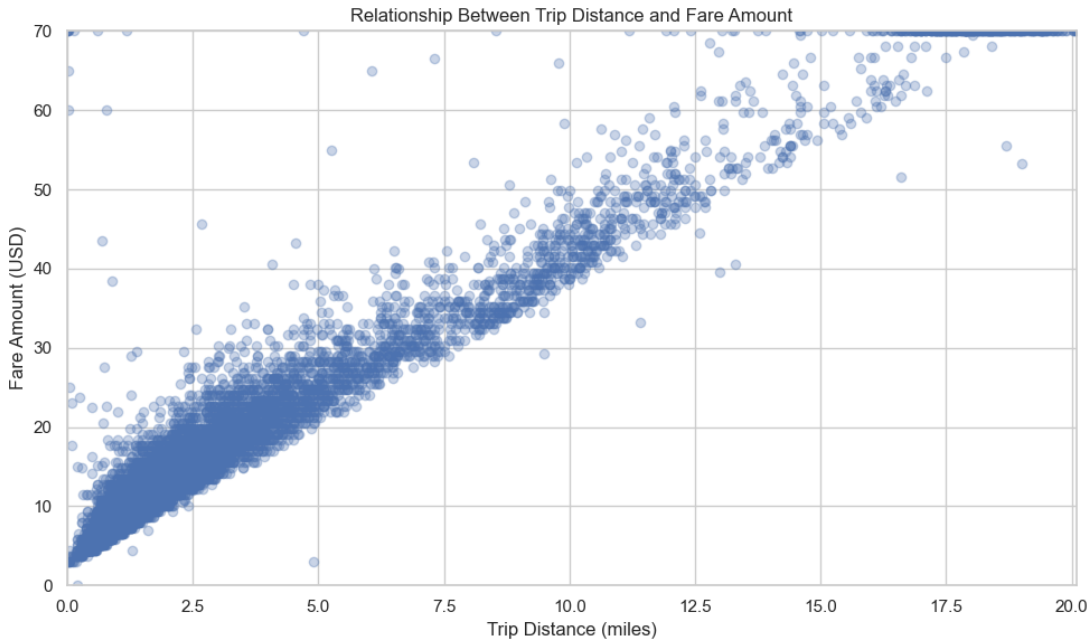

Figure 3: Relationship between trip distance and fare amount. The scatter plot shows a clear positive relationship: longer trips generally correspond to higher fares. Most observations are concentrated below 5 miles, which is consistent with typical short urban taxi travel, while points above the main trend likely reflect tolls, surcharges, or other add-on charges.

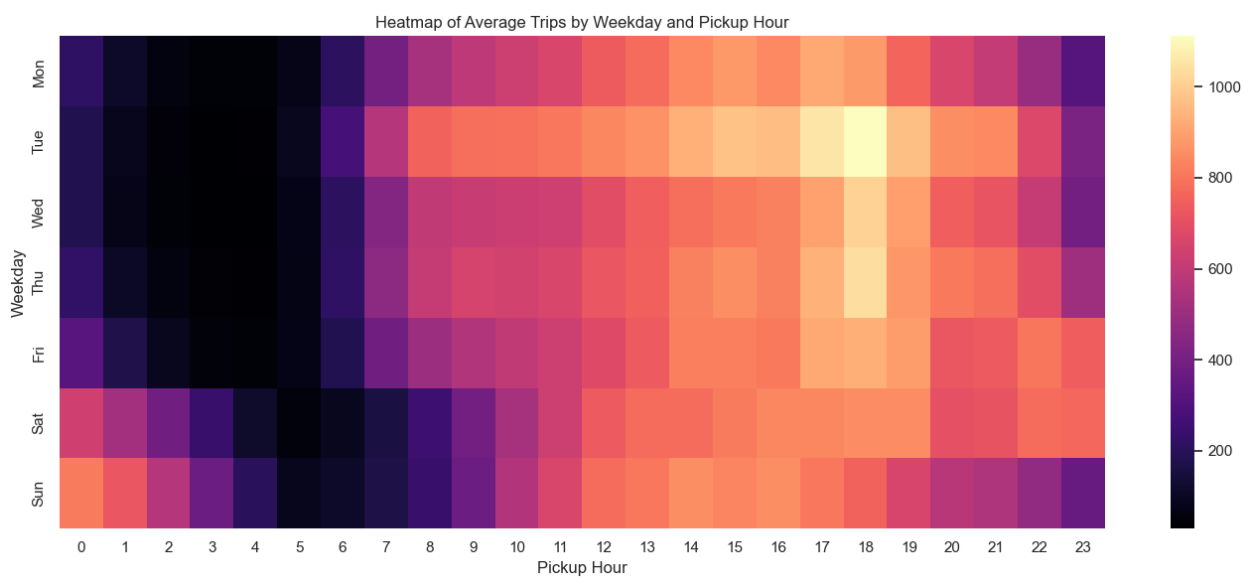

Figure 4: Heatmap of average trips by weekday and pickup hour. Demand is lowest during the early morning hours and increases during daytime and evening periods. Weekday demand is especially strong in the late afternoon and early evening, while weekends show relatively higher late-night activity.

5 Limitations and Additional Data Desired

This analysis is informative but has several limitations.

- The dataset covers only **one month** of trips, so it cannot capture seasonal effects across the year.
- The analysis is limited to variables already present in the public TLC trip data and the lookup table; it does not include weather, public events, transit disruptions, or fuel prices, all of which can influence taxi demand.
- Geographic interpretation is limited by the presence of some unmatched or unknown pickup/dropoff lookup values.
- The tip-rate comparison uses grouped variability rather than a more formal causal model, so it should be interpreted as an association rather than a causal conclusion.

Additional data that would improve the analysis include weather observations, major event calendars, transit service disruptions, airport flight-volume data, and a longer time horizon covering multiple months. These additions would help distinguish routine weekly patterns from one-off external shocks and would support stronger explanations for unusual demand behavior.

6 Conclusion

Using the January 2023 NYC Yellow Taxi dataset, this project built a curated trip-level dataset, applied the required cleaning rules, computed the required groupby and window metrics, answered the three EDA questions, and produced the four required visualizations. The key substantive findings are that the **Bronx** exhibits the strongest weekday effect among NYC boroughs, the most anomalous demand dates are **January 2** and **January 1** during the New Year period, and **tip rate appears more strongly associated with pickup hour than with distance bucket**.

Overall, the results show strong temporal structure in taxi demand and highlight the value of careful cleaning and feature engineering when analyzing large transportation datasets.